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# Acupoint Stimulation After Anesthesia Induction Does Not Decrease Nausea and Vomiting After Hepatectomy: A Multi-Center Double-Blinded Clinical Trial

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## ABSTRACT

**Introduction:** This study evaluated whether adding transcutaneous electrical acupoint stimulation (TEAS) to a double-antiemetic regimen during hepatectomy reduces the incidence of postoperative nausea and vomiting (PONV).

**Methods:** In this multicenter double-blinded randomized controlled trial conducted across three tertiary hospitals in China, patients scheduled for elective hepatectomy were randomized to either the TEAS group (stimulation at PC6 during surgery) or the control group (no stimulation). The primary outcome was the incidence of PONV within 24h postoperatively. Key secondary outcomes included the severity of PONV and pain after surgery.

**Results:** Among 161 randomized patients, 159 completed the study. No significant difference was observed in the incidence of PONV within 24h postoperatively between the TEAS and control groups (23.8% vs. 26.6%; OR, 95% CI: 0.86, 0.42–1.76). However, the TEAS group exhibited less patient controlled analgesia (PCA) attempts (2.7 vs 3.5) or deliveries (2.6 vs 3.5) by 24h postoperatively. Recovery outcomes, including time to ambulation and to gastrointestinal function recovery, did not differ between groups.

**Conclusion:** Intraoperative TEAS did not significantly reduce PONV incidence in hepatectomy patients receiving antiemetic therapy. Nonetheless, its potential to decrease the need for analgesics warrants further investigation to refine its role in perioperative care.

**Clinical trial registration:** ClinicalTrials.gov (NCT05396716), <https://clinicaltrials.gov>

**KEYWORDS** Postoperative nausea and vomiting; transcutaneous electrical acupoint stimulation; hepatectomy; clinical trial

Postoperative nausea and vomiting (PONV) prophylaxis is a key component of enhanced recovery after surgery (ERAS). However, despite the widespread use of double or triple antiemetic therapy, the incidence of PONV remains high in specific patient populations [1]. Epidemiological data from *The Lancet* reveal that China carries the highest global burden of hepatitis B virus (HBV), accounting for nearly one-third of the world's 240 million chronic HBV infections [1]. Without intervention, 20%–30% of chronic hepatitis B cases progress along the clinical spectrum to cirrhosis and/or hepatocellular carcinoma, frequently requiring surgical management [2]. In patients undergoing hepatic resection, PONV may exacerbate hemorrhagic complications and extend hospital stays [1]. Notably, a prospective cohort study reported a PONV incidence of 53.3% following hepatectomy [3].

Recent studies suggest that acupuncture-related therapies, particularly transcutaneous electrical acupoint stimulation (TEAS), may offer perioperative benefits, including PONV mitigation and preservation of gastrointestinal function [4,5]. However, there are research gaps in TEAS being used for

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preventing PONV in the current practice of perioperative management of hepatectomy. First, although TEAS applied at the PC6 (Neiguan) acupoint has demonstrated antiemetic effects in general surgical populations [6], its efficacy specifically in liver surgery remains unverified. Second, standard multimode antiemetic regimen including dexamethasone and 5-HT<sub>3</sub> receptor antagonists are widely used for preventing PONV. The 2015 Cochrane review on PC6 acupoint stimulation for PONV [6] prevention highlighted the importance of determining whether acupoint stimulation can provide further PONV improvement when added to pharmacological prophylaxis. Finally but notably, methodological limitations of existing studies, predominantly unblinded designs raise concerns about expectation bias confounding results [4,6,7]. To address these evidence gaps, we conducted this multicenter double-blind randomized controlled trial to evaluate the effectiveness of intraoperative TEAS under current PONV prophylaxis protocols in hepatic surgery. The work has been reported in line with Consolidated Standards of Reporting Trials (CONSORT) and Transparency In The reporting of Artificial Intelligence (TITAN) Guidelines [8,9].

## METHODS

### *Study Design and Patients*

This multicenter, prospective, randomized clinical trial was conducted in 3 tertiary hospitals and has been approved by the institutional review board of each center (Xijing Hospital, the First Affiliated Hospital of Henan University of Chinese Medicine, Qinghai Provincial People's Hospital). The study was registered at ClinicalTrial.gov (NCT 05396716). Written informed consent was obtained from all participants.

Eligible patients were those who were scheduled for elective hepatectomy under general anesthesia. Exclusion criteria included age younger than 18 years old, body mass index < 18 kg/m<sup>2</sup> or >30 kg/m<sup>2</sup>, American society of anesthesiologists (ASA) status >III, contraindications to TEAS (skin damage or infection at the stimulation site, implanted electronic devices), drug or alcohol abuse, history of neurological or psychiatric disorders, or pregnant/lactating women.

### *Randomization and Blinding*

Participants were randomly allocated (1:1 ratio) to either the TEAS group or the control group through a secure web-based randomization system (Randomiser for Clinical Trial Lite, MEDSHARING). The randomization sequence was generated using permuted blocks, stratified by treatment center. Upon arrival in the operating room, an independent investigator (uninvolved in anesthesia administration or outcome assessment) accessed the randomization system to reveal group assignments. The TEAS intervention commenced immediately after anesthesia induction and continued throughout the surgical procedure. Electrodes were attached to acupoints and connected to a stimulator in both groups. The intervention provider determined the maximum current intensity each patient could tolerate, after which the stimulator was turned off. Immediately following anesthesia induction, the intervention provider adjusted the stimulator settings for patients in the TEAS group and initiated stimulation. For patients in the control group, the intervention provider went through the motions of adjusting the stimulator settings but did not switch the device on. The stimulator was then placed inside an opaque box, preventing operating room personnel from identifying treatment allocation. The patients, anesthesiologists, outcome assessors, and ward staff remained unaware of treatment assignments throughout the study period.

### *Intervention*

The transcutaneous electrical acupoint stimulation (TEAS) procedure strictly adhered to the Standards for Reporting of Controlled Trials in Acupuncture [10]. Surface electrodes were positioned bilaterally at the PC6 acupoints and connected to the stimulator (Hwato SDZ-V; Suzhou Medical Appliances Co., Ltd., China). The device delivered alternating “disperse-dense” wave frequencies (2 Hz and 10 Hz)

in two cycles. Stimulation intensity was individually titrated to each patient's maximum tolerable threshold while maintaining the characteristic "De Qi" sensation (heaviness, numbness, and distension at the stimulation site). In the TEAS group, bilateral acupoint stimulation was administered from immediately after anesthesia induction until surgery completion. Immediately after anesthesia induction was defined as loss of consciousness after administration of propofol and sufentanil. For control group participants, electrodes were similarly placed but remained inactive.

## Procedures

Standard intraoperative monitoring was implemented, including continuous electrocardiography, pulse oximetry, invasive arterial blood pressure monitoring, and Narcotrend index measurement for depth of anesthesia assessment. Anesthesia induction was achieved using midazolam (0.03–0.05 mg/kg), sufentanil (0.3–0.5 µg/kg), and propofol (1.5–2.5 mg/kg), with rocuronium (0.6 mg/kg) administered to facilitate endotracheal intubation. Anesthesia maintenance consisted of sevoflurane (1–1.5 age-adjusted MAC), remifentanyl infusion (0.1–0.3 µg/kg/min), and dexmedetomidine (0.4 µg/kg/h), with supplemental rocuronium boluses as required. Preemptive analgesia was provided with parecoxib sodium (40 mg IV) and dexamethasone (10 mg IV) prior to surgical incision. The Narcotrend index was maintained between 40 and 60 throughout the procedure. Antiemetic prophylaxis included palonosetron (0.25 mg IV), with oxycodone (5 mg IV) administered for postoperative pain control prior to surgical completion. Following extubation, patients were transferred to the post-anesthesia care unit (PACU) for recovery. Postoperative analgesia was maintained using an intravenous patient-controlled analgesia (PCA) system programmed to deliver a solution containing sufentanil (100 µg) and oxycodone (20 mg) in normal saline, with a background infusion rate of 0.015 ml/kg/h, bolus doses of 0.015 ml/kg, and a 10-minute lockout interval.

## Outcomes

The primary outcome was the incidence of PONV by 24 h postoperatively. Secondary outcomes included visual analogue scale (VAS) score of PONV, nausea and vomiting during PACU stay, by 24 h and 48 h postoperatively; incidence of PONV during PACU stay and by 48 h postoperatively; the duration between the end of surgery and the onset of nausea and vomiting; the VAS of pain during PACU stay, by 24 h and 48 h postoperatively; the number of total PCA attempts and effective PCA deliveries by 24 h and 48 h after surgery; the 15-item quality of recovery (QoR-15) score by 24 h and 48 h postoperatively; length of hospital stay after surgery; time to ambulation, to flatus and to oral intake after surgery; major complications during hospitalization; and mortality at 6 months postoperatively.

## Statistical Analysis

The primary endpoint was the incidence of PONV by 24 h postoperatively. According to a previous report [11], we anticipated that the incidence of PONV in patients undergoing hepatectomy was 55%. Assuming a reduction of incidence of PONV to 33% after TEAS based on our pilot study, to detect this reduction with an  $\alpha$  of 0.05 (two-sided) and 80% power ( $\beta=0.2$ ), we required 72 patients per group. Accounting for a potential 10% attrition rate, we enrolled a total of 158 participants (79 per group) to ensure adequate statistical power.

Data analyses were primarily performed in a modified intention-to-treat population, which included all randomly assigned participants who initiated the study intervention and did not withdraw consent. The primary endpoint was compared with an unadjusted  $\chi^2$  test and results are presented as relative risks (odds ratio, OR) with 95% confidence intervals. Results for the primary outcome are additionally reported as absolute risk difference (RD) with 95% confidence intervals. Pre-defined subgroup analyses were done based on gender, history of smoking, motion sickness, chemotherapy and type of surgery (laparoscopic surgery or open surgery).

For secondary endpoints, OR and RD together with a 95% CI were calculated using a general linear model for binomial outcomes. Scores were compared between groups with ordered logistic regression. Time-to-event results (time to flatus, to oral intake of water, to ambulation and to discharge) were analyzed with Kaplan–Meier survival analyses with log-rank tests. Univariable Cox proportional hazards models were used to calculate the hazard ratio and 95% CI.

Given that fewer than 5% of data points were missing, no imputation methods were applied. We considered secondary outcomes as exploratory and thus did not include any multiplicity adjustment.

Post-hoc sensitivity analyses included analysis of the primary outcome in the per-protocol (PP) population, analysis on the incidence of nausea and vomiting respectively. Mixed effects logistic regression analysis by dose of opioid and type of disease as fixed covariates and trial center as a random effect were performed as well [12].

Statistical significance was defined as a two-tailed  $p$  value  $<0.05$ . The statistical analysis was conducted using SPSS 25.0 software package (IBM SPSS Inc., Chicago, IL, USA) and R Studio 4.3.0 (R Studio, Boston, MA, USA). Details of the statistical analysis plan are shown in [Supplemental Material 1](#).

## RESULTS

### *Patient Characteristics*

Between June 5, 2022, and May 17, 2023, a total of 201 patients were screened, of whom 161 were randomly assigned to group TEAS and group Control. The last follow-up was November 17, 2023. One patient in group TEAS had surgery canceled and one patient in group Control withdrew the informed consent, both were excluded from the mITT analysis. Four patients in group TEAS and two patients in group Control were excluded from PP analysis due to a variation to the protocol. Thus, the PP set included 76 patients in group TEAS and 77 patients in group Control ([Figure 1](#)). The mean (SD) age of participants was 53.3 (9.2) years; 54 patients were women (34.0%). Race and ethnicity data were not collected. The two groups were well balanced at baseline ([Table 1](#)). Baseline characteristics of the PP population are shown in [Supplemental Material 2](#) (eTable 2).

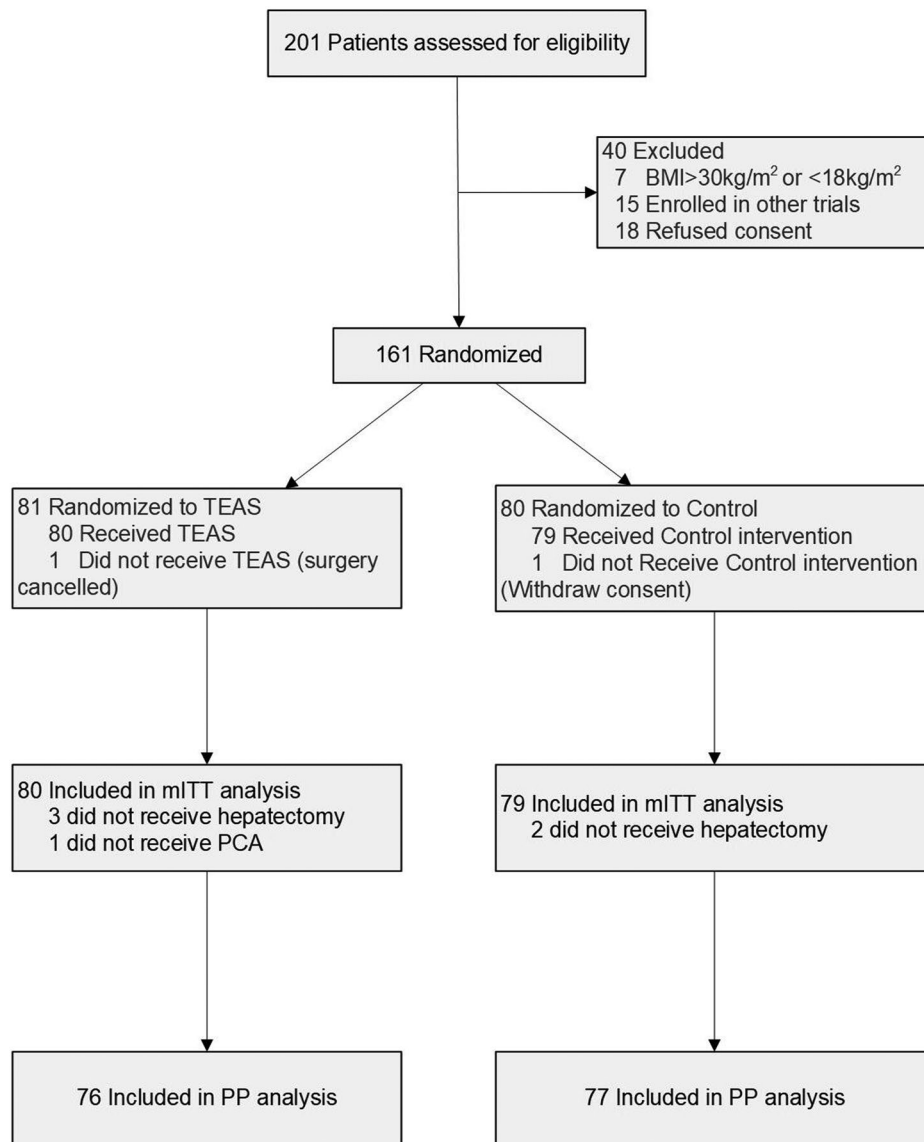
### *Primary Outcomes*

For the mITT analysis, the incidence of PONV by 24 h after surgery was 23.8% in the TEAS group and 26.6% in the control group (OR = 0.86, 95% CI, 0.42–1.76;  $p=0.68$ ). Based on the significance level of .05, there was no statistical evidence of a difference between the two groups. Findings were consistent for the per-protocol population (OR = 0.89, 95% CI, 0.43–1.83;  $p=0.75$ ) ([Table 2](#)).

### *Secondary Outcomes*

Using mITT analysis, no statistically significant difference was detected for the scores of PONV assessed with PONV scale and VAS score by 24h and 48h after surgery ([Table 3](#)). For nausea VAS score during PACU stay, a statistically significant difference was detected in favor of TEAS (median, range: 0, 2 vs 0, 3;  $p=0.046$ ). Analysis of pain VAS scores revealed no statistically significant inter-group differences at any postoperative assessment time point ([Table 3](#)). However, compared with the Control group, the TEAS group showed less total PCA attempts ( $2.7 \pm 3.3$  vs  $3.5 \pm 6.1$ ,  $p=0.02$ ) and less effective deliveries ( $2.6 \pm 3.3$  vs  $3.5 \pm 6.0$ ,  $p=0.02$ ) by 24h after surgery. Total PCA attempts and effective deliveries were not different between the two groups by 48h after surgery. Moreover, the total opioid consumption on morphine equivalent by 24h and 48h after surgery were not different between the two groups ([Supplemental material 2](#), eTable 1). No statistically significant difference in QoR-15 was detected between treatment arms ([Table 3](#)).

Analysis of time to flatus, to oral intake and to ambulation after surgery identified no statistically significant differences between the two groups ([Figure 2](#)). Although two patients in the TEAS group died within 6 months after surgery compared with none in the control group, the difference in mortality rates between groups was not statistically significant ( $p=0.1$ , [Table 2](#)).



**FIGURE 1.** CONSORT flow diagram of patient enrollment, randomization, and follow-up. BMI: body mass index; TEAS: transcutaneous electrical acupoint stimulation; mITT: modified intention-to-treat; PP: per-protocol; PCA: patient-controlled analgesia.

### Subgroup Analysis and Sensitivity Analysis

Results of the primary endpoint were generally consistent across predefined subgroups by gender, chemotherapy, motion sickness, smoking and type of surgery (laparoscopic or open surgery), with no statistically significant interactions (Figure 3). Mixed effects logistic regression analysis by dose of opioid and type of disease as fixed covariates and trial center as a random effect showed no significant difference in the incidence of PONV by 24h after surgery (Table 2).

## DISCUSSION

Our results show no additional benefit by the use of intraoperative TEAS to the standard dual anti-emetic regimen in the prevention of PONV after hepatectomy. The incidences of PONV were not different between the groups. As for the severity of PONV, though the VAS scores of nausea in PACU were statistically different, it lacks clinical relevance (0 vs 0). To our knowledge, this is the first



**TABLE 1.** Baseline characteristics of study population.

| Characteristic                             | TEAS (n=80)        | Control (n=79)     |
|--------------------------------------------|--------------------|--------------------|
| Sex, No. (%)                               |                    |                    |
| Female                                     | 28 (35.0)          | 26 (32.9)          |
| Male                                       | 52 (65.0)          | 53 (67.1)          |
| Age, mean $\pm$ SD, yrs                    | 52.3 $\pm$ 9.1     | 54.3 $\pm$ 10.9    |
| BMI, mean $\pm$ SD, kg/m <sup>2</sup>      | 23.4 $\pm$ 3.3     | 23.7 $\pm$ 4.9     |
| ASA grade, No. (%)                         |                    |                    |
| I                                          | 2 (2.5)            | 1 (1.3)            |
| II                                         | 74 (92.5)          | 72 (91.1)          |
| III                                        | 4 (5)              | 6 (7.6)            |
| Smoking, No. (%)                           | 25 (31.3)          | 23 (29.1)          |
| Chemotherapy, No. (%)                      | 28 (35.0)          | 24 (30.4)          |
| Motion sickness, No. (%)                   | 15 (18.8)          | 11 (13.9)          |
| Hypertension, No. (%)                      | 12 (15.0)          | 9 (11.4)           |
| Preoperative nausea/vomiting, No. (%)      | 0 (0)              | 0 (0)              |
| History of PONV, No. (%)                   | 0 (0)              | 0 (0)              |
| Diagnosis, No. (%)                         |                    |                    |
| Liver cancer                               | 62 (77.5)          | 57 (72.2)          |
| Bile duct cancer                           | 3 (3.8)            | 9 (11.4)           |
| Hemangioma                                 | 10 (12.5)          | 6 (7.6)            |
| Bile duct stones                           | 1 (1.3)            | 4 (5.1)            |
| Liver echinococcosis                       | 4 (5)              | 3 (3.8)            |
| Type of surgery, No. (%)                   |                    |                    |
| Laparoscopic surgery                       | 39 (48.8)          | 35 (44.3)          |
| Open abdominal surgery                     | 41 (51.3)          | 44 (55.7)          |
| Length of surgery, mean $\pm$ SD, min      | 221.1 $\pm$ 97.6   | 245.3 $\pm$ 108.9  |
| Length of anesthesia, mean $\pm$ SD, min   | 276.1 $\pm$ 129.2  | 293.3 $\pm$ 130.2  |
| Dose of sufentanil, mean $\pm$ SD, $\mu$ g | 32.1 $\pm$ 8.2     | 32.3 $\pm$ 7.2     |
| Dose of remifentanyl, mean $\pm$ SD, mg    | 2.5 $\pm$ 1.4      | 2.8 $\pm$ 1.9      |
| Input of colloid, mean $\pm$ SD, mL        | 751.9 $\pm$ 703.8  | 670.9 $\pm$ 413.0  |
| Input of crystalloid, mean $\pm$ SD, mL    | 1824.6 $\pm$ 729.9 | 1873.9 $\pm$ 707.2 |
| Bleeding, mean $\pm$ SD, mL                | 653.0 $\pm$ 979.1  | 488.1 $\pm$ 514.9  |

TEAS: transcutaneous electrical acupoint stimulation; BMI: body mass index; ASA: American Society of Anesthesiologists; SD: standard deviation.

**TABLE 2.** Primary outcome.

|                 | TEAS      | Control   | RD (95%CI)          | Or (95%CI) adjusted or (95% CI)*       | p值           |
|-----------------|-----------|-----------|---------------------|----------------------------------------|--------------|
| mITT            |           |           |                     |                                        |              |
| No. of patients | 80        | 79        |                     |                                        |              |
| PONV, No. (%)   | 19 (23.8) | 21 (26.6) | -0.03 (-0.16, 0.11) | 0.86 (0.42, 1.76)<br>0.84 (0.38, 1.70) | 0.68<br>0.62 |
| PP              |           |           |                     |                                        |              |
| No. of patients | 76        | 77        |                     |                                        |              |
| PONV, No. (%)   | 19 (25.0) | 21 (27.3) | -0.03 (-0.15, 0.12) | 0.89 (0.43, 1.83)<br>0.87 (0.39, 1.78) | 0.75<br>0.74 |

TEAS: transcutaneous electrical acupoint stimulation; mITT: modified intention-to-treat; PP: per-protocol; RD: risk difference; OR: odds ratio; CI: confidence interval; PONV: postoperative nausea and vomiting.

\*Estimated from a mixed effects logistic regression model, adjusted for dose of opioid and type of disease as a fixed covariates and trial center as a random effect.

double-blind clinical trial on the effect of TEAS on PONV in patients undergoing abdominal surgery. Though TEAS has been accepted as part of multimodal antiemetic prophylaxis in high-risk patients undergoing various surgeries [13], our results indicate that an appropriate protocol may be important for the application of TEAS.

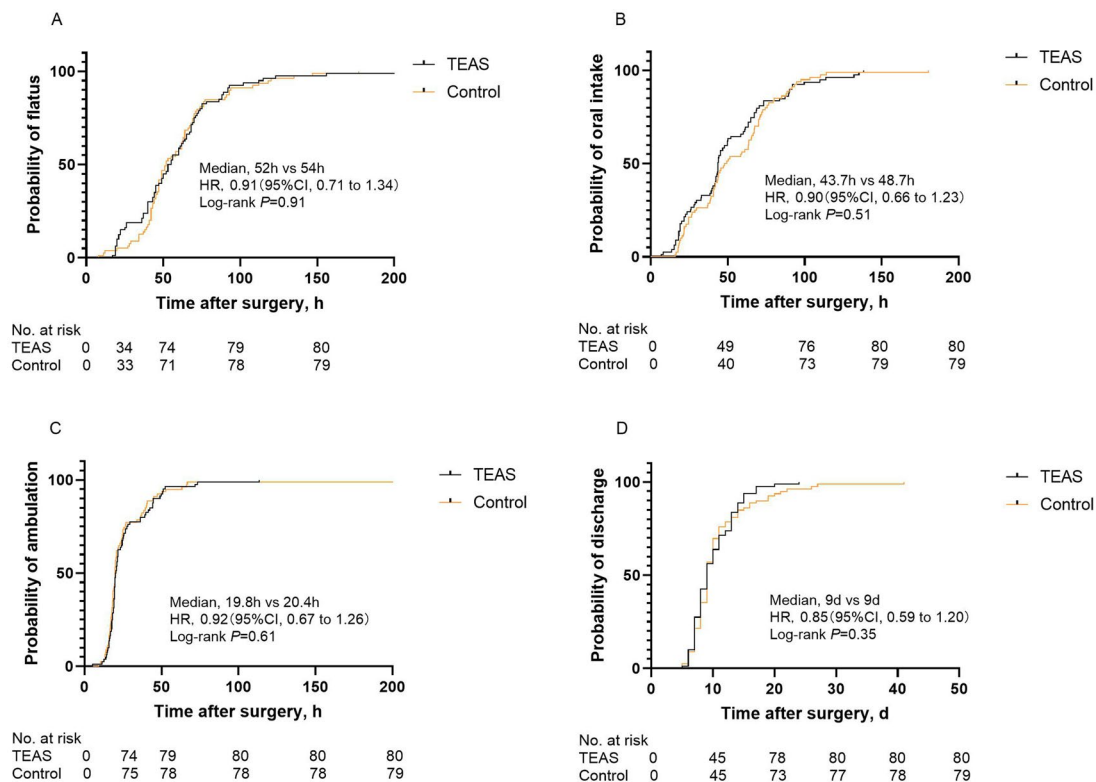
The first reason for the negative results in our study may be the insufficient amount of acupoint stimulation. Compared with single intraoperative acupoint stimulation, a multi-session stimulation protocol administered preoperatively, intraoperatively, and postoperatively might be more effective in reducing PONV. Zhang et al. conducted a non-inferiority trial comparing TEAS with dexamethasone for PONV prevention. Their results confirmed that only the stimulation protocol administered '0.5 h before anesthesia induction, immediately after entering the post-anesthesia care unit, and every 3 h after leaving the post-anesthesia care unit' could achieve non-inferiority to 8 mg dexamethasone [14]. Another study in gynecological laparoscopic surgery reported that multiple-period TEAS provides greater efficacy and a longer duration of prophylactic antiemetic effect than single-period TEAS as well [15]. Moreover, a combination of different acupoint stimulation techniques, including TEAS,

**TABLE 3.** Secondary outcomes.

| Characteristics          | TEAS(n=80) | Control(n=79) | p value |
|--------------------------|------------|---------------|---------|
| During PACU stay         |            |               |         |
| VAS score of PONV        | 0 (2)      | 0 (3)         | 0.12    |
| Scale of Nausea          | 0 (2)      | 0 (3)         | 0.046   |
| Scale of vomiting        | 0 (1)      | 0 (1)         | 0.68    |
| VAS score of pain        | 1 (6)      | 1 (7)         | 0.39    |
| 24h after surgery        |            |               |         |
| VAS score of PONV        | 0 (6)      | 0 (8)         | 0.82    |
| Scale of Nausea          | 0 (3)      | 0 (4)         | 0.96    |
| Scale of vomiting        | 0 (3)      | 0 (3)         | 0.64    |
| VAS score of pain        | 3 (8)      | 3 (9)         | 0.34    |
| Total PCA attempts       | 2.7 ± 3.3  | 3.5 ± 6.1     | 0.02    |
| Effective PCA deliveries | 2.6 ± 3.3  | 3.5 ± 6.0     | 0.02    |
| QoR-15 score             | 105 (86)   | 105 (134)     | 0.93    |
| 48h after surgery        |            |               |         |
| VAS score of PONV        | 0 (5)      | 0 (5)         | 0.42    |
| Scale of Nausea          | 0 (2)      | 0 (2)         | 0.42    |
| Scale of vomiting        | 0 (1)      | 0 (2)         | 0.99    |
| VAS score of pain        | 2 (8)      | 3 (8)         | 0.17    |
| Total PCA attempts       | 4.7 ± 5.3  | 6.1 ± 9.4     | 0.05    |
| Effective PCA deliveries | 4.8 ± 5.5  | 6.4 ± 10.9    | 0.05    |
| QoR-15 score             | 118 (75)   | 116 (80)      | 0.69    |
| 6m after surgery         |            |               |         |
| Death                    | 2 (2.5)    | 0 (0)         | 0.1     |

Data were presented as mean ± standard deviation, median (range) or number (percentage).

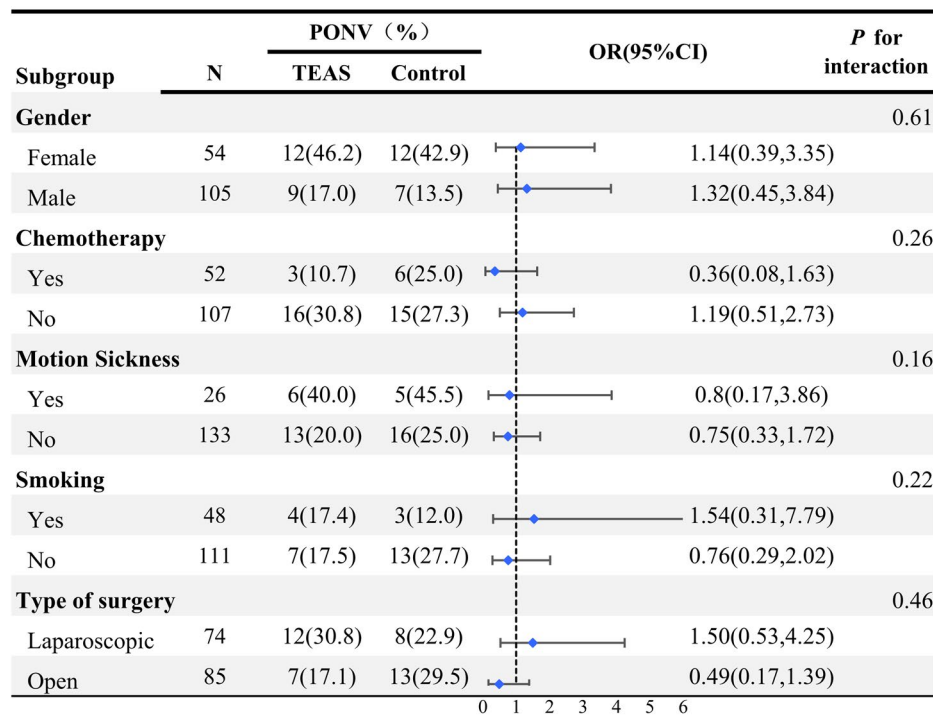
TEAS: transcutaneous electrical acupoint stimulation; PACU: post-anesthesia care unit; VAS: visual analogue scale; PCA: patient-controlled analgesia; QoR-15: the 15-item quality of recovery.



**FIGURE 2.** Kaplan–Meier estimates of the probability of failure to flatus, oral intake, or ambulation by discharge. The TEAS group is represented by the black line, and the control group by the orange line. Groups were compared using the log-rank test. TEAS: transcutaneous electrical acupoint stimulation.

acupressure, and electroacupuncture, may be more effective [16]. Based on our results and the previous studies, we speculate that perioperative TEAS requires a certain cumulative dosage and treatment duration to achieve optimal antiemetic effects. Multiple TEAS administrations during and after surgery may be more beneficial for PONV prevention and treatment.





**FIGURE 3.** Forest Plot of subgroup analysis of the incidence of PONV by 24h after surgery. Subgroup analyses were performed according to gender, chemotherapy, motion sickness, smoking, and type of surgery. For each study, the blue dot indicates the OR, and the horizontal line indicates the 95% confidence interval. The *p* value for interaction between subgroups was all over 0.05, suggesting that the stratified factors did not significantly modify the effect of TEAS on PONV. PONV: postoperative nausea and vomiting; TEAS: transcutaneous electrical acupoint stimulation; OR: odds ratio; CI: confidence interval.

Other potential reasons why a treatment effect was not seen with TEAS may be that all participants received standard dual pharmacologic prophylaxis with dexamethasone and 5-HT<sub>3</sub> receptor antagonists. Moreover, the actual PONV incidence in the control group was only 26.6%, much lower than the anticipated 55%. Under such circumstances, a single intraoperative TEAS administration failed to demonstrate additional benefits when added to this standardized multimodal antiemetic protocol. Comparative studies demonstrate that transcutaneous electrical acupoint stimulation (TEAS) enhances PONV prophylaxis when combined with antiemetics, though its efficacy depends on baseline PONV risk. In laparoscopic sleeve gastrectomy [13], TEAS added to dexamethasone/tropisetron reduced PONV incidence from 77.4% to 41.9% (45.9% relative reduction), while in gynecologic laparoscopy [14], TEAS plus dexamethasone achieved 24-h PONV rates of 28% versus 50% with dexamethasone alone (OR 0.39, 95% CI 0.17–0.89), comparable to tropisetron plus dexamethasone (26%, OR 0.36, 95% CI 0.16–0.82). However, when using dexamethasone/palonosetron (current study), the PONV incidence of group Control was only 26.6%, suggesting that TEAS's additive effect may be obscured in low-risk settings where modern antiemetics already achieve substantial PONV reduction. Notably, TEAS showed non-inferiority to antiemetics (*p*=0.857 vs. tropisetron) and has been successfully implemented in breast surgery [15], supporting its role as a viable non-pharmacologic adjunct, particularly in moderate-to-high-risk populations where baseline PONV rates exceed 50%.

A particular feature of this trial was the beginning of the intervention after anesthesia induction, which ascertained the blinding of the participants to the group allocation. The issue of PONV management by TEAS has been debated over the past years, a significant component of the controversy being the unblinding of participants during stimulation. Previous studies have described themselves as double-blind, but it can be observed that the blinding effect on subjects in their study designs was not actually achieved. For example, Jin's study [16] reported being double-blind, yet the intervention was administered postoperatively in the ward, making it unlikely for subjects to remain

unaware of their group assignment. Moreover, such questionable blinding methods are remarkably common in perioperative acupoint stimulation research. In clinical trials investigating the sedative and analgesic effects of acupoint stimulation, the interventions are frequently administered while patients remain conscious [15,17,18], making it impossible to guarantee subjects' unawareness of their group allocation. Only interventions initiated after anesthesia induction can truly ensure proper blinding of study participants.

The acupoints stimulated in our study are bilateral PC6. PC6 is the most frequently chosen acupoint for the prevention and treatment of PONV [19]. However, compared with PC6 alone, concomitant stimulation of the PC6 and the L14 acupuncture points may induce a more prophylactic antiemetic effect after general anesthesia [20]. Furthermore, to achieve other therapeutic effects, the selection of different acupoints may be necessary. For instance, the Zusanli acupoint (ST36) is typically chosen for preventing postoperative ileus [4,8,21–23]. This specificity in acupoint selection might also partially explain why our study failed to observe TEAS effects on time to first flatus, time to oral intake, and time to ambulation.

We observed less boluses of PCA, which is in accordance with the previous studies [23,24]. The beneficial analgesic effect of TEAS might depend on the stimulation of PC6 which is frequently used in studies on postoperative analgesia [25]. However, the total opioid consumption of PCA was not different between the groups, indicating that the analgesic effect of TEAS is relatively weak.

### Limitations

This study is not without limitations. First, we acknowledge that the lower-than-anticipated PONV incidence (26.6% in controls vs. 40% assumed in sample size calculation) may have reduced the study's power. A post-hoc power analysis revealed that with the observed rates (23.8% vs. 26.6%) and  $N=160$ , the power to detect a 40% relative risk reduction was only 27.7% (using  $\alpha=0.05$ , two-sided test), indicating insufficient statistical power. The non-significant result should therefore be interpreted as inconclusive rather than definitive evidence of no effect. Future trials with larger samples, tailored to the actual PONV incidence, are needed to confirm or refute our findings. Second, the study population predominantly consisted of male participants, while female gender is an independent risk factor for PONV. However, our predefined subgroup analysis demonstrated no interaction effect between gender and intervention groups, indicating that gender did not influence TEAS efficacy. Third, given that our study employed a single-session TEAS protocol confined to the period between anesthesia induction and surgical conclusion, both the stimulation duration and cumulative dose were inherently limited. This restricted application may account for the observed ineffectiveness of TEAS in preventing PONV, as the intervention might not have reached its full therapeutic potential under these constrained conditions. Additionally, although we observed a statistically significant difference in nausea scores between the TEAS and control groups in the PACU, the actual difference (0 vs 0) lacked clinical significance.

### CONCLUSION

In summary, our study demonstrates that additional intraoperative TEAS administered at PC6 acupoints does not further reduce the incidence of PONV in patients undergoing hepatectomy when added to the current antiemetic regimen. However, this TEAS protocol may potentially decrease postoperative analgesic requirements.

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## AUTHOR CONTRIBUTIONS

CRediT: **Shan He**: Conceptualization, Investigation, Writing – original draft; **Zhifeng Lv**: Conceptualization, Investigation, Resources; **Hongxia Wang**: Conceptualization, Investigation; **Nannan Yu**: Methodology; **Xiaoshuang Tuo**: Formal analysis; **Yun Wang**: Conceptualization, Investigation, Writing – review & editing, Resources, Supervision; **Hailong Dong**: Supervision, Writing – review & editing; **Jingyi Cao**: Writing – original draft; **Yonghui Wang**: Conceptualization, Investigation, Writing – review & editing, Resources, Supervision; **Zhihong Lu**: Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

## DISCLOSURE STATEMENT

No potential conflict of interest was reported by the author(s).

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## DATA AVAILABILITY STATEMENT

The deidentified individual participant data supporting the findings and results of this study will be made available after publication upon personal request to Dr. Zhihong Lu ([deerlu23@163.com](mailto:deerlu23@163.com)).

## REFERENCES

1. Darvall JN, Leslie K. Pounds of prevention but only ounces of cure: the need for more research on the treatment of postoperative nausea and vomiting. *Anesthesiology*. 2019;130(2):183–185. doi:[10.1097/aln.0000000000002536](https://doi.org/10.1097/aln.0000000000002536).
2. Chen S, Li J, Wang D, et al. The hepatitis B epidemic in China should receive more attention. *Lancet*. 2018;391(10130):1572. doi:[10.1016/s0140-6736\(18\)30499-9](https://doi.org/10.1016/s0140-6736(18)30499-9).
3. Chen X, Zhou H, Gao J, et al. External validation of postoperative nausea and vomiting risk scores in patients with liver cancer: a single-centre prospective cohort study. *Eur J Oncol Nurs*. 2023;65:102350. doi:[10.1016/j.ejon.2023.102350](https://doi.org/10.1016/j.ejon.2023.102350).
4. Liu Y, May BH, Zhang AL, et al. Acupuncture and related therapies for treatment of postoperative ileus in colorectal cancer: a systematic review and meta-analysis of randomized controlled trials. *Evid Based Complement Alternat Med*. 2018;2018(1):3178472. doi:[10.1155/2018/3178472](https://doi.org/10.1155/2018/3178472).
5. Song G, Trujillo S, Fu Y, et al. Transcutaneous electrical stimulation for gastrointestinal motility disorders. *Neurogastroenterol Motil*. 2023;35(11):e14618. doi:[10.1111/nmo.14618](https://doi.org/10.1111/nmo.14618).
6. Lee A, Zhang JZ, Xie J, Cheng V, Wong MKH, Yau DKW. Stimulation of the wrist acupuncture point PC6 for preventing postoperative nausea and vomiting. *Cochrane Database Syst Rev*. 2025;9(9):CD003281. doi:[10.1002/14651858.CD003281.pub4](https://doi.org/10.1002/14651858.CD003281.pub4).
7. Wang Y, Yang JW, Yan SY, et al. Electroacupuncture vs sham electroacupuncture in the treatment of postoperative ileus after laparoscopic surgery for colorectal cancer: a multicenter, randomized clinical trial. *JAMA Surg*. 2023;158(1):20–27. doi:[10.1001/jamasurg.2022.5674](https://doi.org/10.1001/jamasurg.2022.5674).
8. Hopewell S, Chan AW, Collins GS, et al. CONSORT 2025 statement: updated guideline for reporting randomised trials. *BMJ*. 2025;389:e081123. doi:[10.1136/bmj-2024-081123](https://doi.org/10.1136/bmj-2024-081123).
9. Agha RA, Mathew G, Rashid R, et al. Transparency in the reporting of artificial intelligence – the TITAN guideline. *Premier J Sci*. 2025;10:100082. doi:[10.70389/PJS.100082](https://doi.org/10.70389/PJS.100082).
10. MacPherson H, Altman DG, Hammerschlag R, et al. Revised STAndards for reporting interventions in clinical trials of acupuncture (STRICTA): extending the CONSORT statement. *Acupunct Med*. 2010;28(2):83–93. doi:[10.1136/aim.2009.001370](https://doi.org/10.1136/aim.2009.001370).
11. Sun JX, Bai KY, Liu YF, et al. Effect of local wound infiltration with ropivacaine on postoperative pain relief and stress response reduction after open hepatectomy. *World J Gastroenterol*. 2017;23(36):6733–6740. doi:[10.3748/wjg.v23.i36.6733](https://doi.org/10.3748/wjg.v23.i36.6733).
12. Cao SJ, Zhang Y, Zhang YX, et al. Delirium in older patients given propofol or sevoflurane anesthesia for major cancer surgery: a multicentre randomised trial. *Br J Anaesth*. 2023;131(2):253–265. doi:[10.1016/j.bja.2023.04.024](https://doi.org/10.1016/j.bja.2023.04.024).
13. Xiong Q, Min S, Wei K, et al. Transcutaneous electrical acupoint stimulation combined with dexamethasone and tropisetron prevents postoperative nausea and vomiting in female patients undergoing laparoscopic

- sleeve gastrectomy: a prospective, randomized controlled trial. *Obes Surg*. 2021;31(5):1912–1920. doi:[10.1007/s11695-020-05205-9](https://doi.org/10.1007/s11695-020-05205-9).
14. Yang XY, Xiao J, Chen YH, et al. Dexamethasone alone vs in combination with transcutaneous electrical acupoint stimulation or tropisetron for prevention of postoperative nausea and vomiting in gynaecological patients undergoing laparoscopic surgery. *Br J Anaesth*. 2015;115(6):883–889. doi:[10.1093/bja/aev352](https://doi.org/10.1093/bja/aev352).
  15. Zhang Y, Li Y, Ji F, et al. Transcutaneous electrical acupoint stimulation versus dexamethasone for prophylaxis of postoperative nausea and vomiting in breast surgery: a non-inferiority randomized controlled trial. *Surgery*. 2023;174(4):787–793. doi:[10.1016/j.surg.2023.06.014](https://doi.org/10.1016/j.surg.2023.06.014).
  16. Jin LD, Lei W, Xu J, et al. Effect of single or multi-period use of transcutaneous acupoint electrical stimulation on postoperative nausea and vomiting in patients undergoing gynecological laparoscopic surgery: a prospective randomized double-blind trial. *BMC Complement Med Ther*. 2025;25(1):110. doi:[10.1186/s12906-025-04847-5](https://doi.org/10.1186/s12906-025-04847-5).
  17. Sun R, Dai W, Liu Y, et al. Non-needle acupoint stimulation for prevention of nausea and vomiting after breast surgery: a meta-analysis. *Medicine*. 2019;98(10):e14713. doi:[10.1097/md.00000000000014713](https://doi.org/10.1097/md.00000000000014713).
  18. Chen J, Zhang Y, Li X, et al. Efficacy of transcutaneous electrical acupoint stimulation combined with general anesthesia for sedation and postoperative analgesia in minimally invasive lung cancer surgery: a randomized, double-blind, placebo-controlled trial. *Thorac Cancer*. 2020;11(4):928–934. doi:[10.1111/1759-7714.13343](https://doi.org/10.1111/1759-7714.13343).
  19. Qin J, Ye X, Ye C, et al. The effect of transcutaneous electrical acupoint stimulation on high-risk patients with PONV undergoing laparoscopic gynecologic surgery: a randomized controlled trial. *J Clin Med*. 2023;12(3):1192. doi:[10.3390/jcm12031192](https://doi.org/10.3390/jcm12031192).
  20. Alizadeh R, Esmaili S, Shoar S, et al. Acupuncture in preventing postoperative nausea and vomiting: efficacy of two acupuncture points versus a single one. *J Acupunct Meridian Stud*. 2014;7(2):71–75. doi:[10.1016/j.jams.2013.04.005](https://doi.org/10.1016/j.jams.2013.04.005).
  21. Yang JW, Shao JK, Wang Y, et al. Effect of acupuncture on postoperative ileus after laparoscopic elective colorectal surgery: a prospective, randomised, controlled trial. *EClinicalMedicine*. 2022;49:101472. doi:[10.1016/j.eclinm.2022.101472](https://doi.org/10.1016/j.eclinm.2022.101472).
  22. Zhao X, Si S, Liu X, et al. Does invasive acupuncture improve postoperative ileus after colorectal cancer surgery? A systematic review and meta-analysis. *Front Med*. 2023;10:1201769. doi:[10.3389/fmed.2023.1201769](https://doi.org/10.3389/fmed.2023.1201769).
  23. Lu Z, Dong H, Wang Q, et al. Perioperative acupuncture modulation: more than anesthesia. *Br J Anaesth*. 2015;115(2):183–193. doi:[10.1093/bja/aev227](https://doi.org/10.1093/bja/aev227).
  24. Lu Z, Wang Q, Sun X, et al. Transcutaneous electrical acupoint stimulation before surgery reduces chronic pain after mastectomy: a randomized clinical trial. *J Clin Anesth*. 2021;74:110453. doi:[10.1016/j.jclinane.2021.110453](https://doi.org/10.1016/j.jclinane.2021.110453).
  25. Shah S, Godhardt L, Spofford C. Acupuncture and postoperative pain reduction. *Curr Pain Headache Rep*. 2022;26(6):453–458. doi:[10.1007/s11916-022-01048-4](https://doi.org/10.1007/s11916-022-01048-4).